

WILMS

Women's Interest-Free Loan Management System
Operations Manual — v1.7.3

WILMS Operations Manual

Version: 1.7.3 Audience: DevOps, Super Admins, programme IT leads Classification: Confidential

1. Purpose

This manual covers day-to-day operational procedures for running WILMS in production and staging environments.

2. Environment overview

Environment	URL	Purpose
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Production	wilms.vercel.app	Live programme operations
Preview	Vercel preview deployments	PR validation
Local	localhost:3000	Development

3. Health monitoring

Health check

GET /api/wilms/health

Returns service status, version, database connectivity.

Ops dashboard

Super Admin accesses /ops for:

- Active incidents
- Maintenance windows
- System metrics summary

4. Daily operations checklist

Task	Frequency	Owner
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Verify health endpoint	Daily	DevOps
Review ops incidents	Daily	Super Admin

| Monitor reconciliation aging | Daily | Super Admin |

| Review notification cron logs | Daily | DevOps |

| Check Neon connection pool | Weekly | DevOps |

| Review audit log anomalies | Weekly | Auditor/Admin |

| Run smoke tests post-deploy | Per deploy | DevOps |

5. Deployment procedure

Pre-deploy

1. Verify branch passes CI (lint, type-check, test). 2. Run `npm run verify:version`. 3. Run `npm run verify:migrations`. 4. Review CHANGELOG for release notes.

Deploy

1. Merge to production branch. 2. Vercel auto-deploys on push. 3. Migrations applied: `npm run db:migrate -w @wilms/domain`. 4. Post-deploy smoke: `npm run smoke:production`.

Rollback

1. Revert Vercel deployment to previous build. 2. Database rollback requires manual SQL — migrations are forward-only. 3. Document incident in ops module.

6. Database operations

Connection

Neon serverless PostgreSQL via `DATABASE_URL`. Connection pooling enabled.

Migrations

`npm run db:migrate -w @wilms/domain` `npm run verify:migrations -w @wilms/api`

Backup

Neon provides automated point-in-time recovery. Verify backup retention in Neon dashboard. Run restore drill: `npm run drill:backup-restore`.

Demo data cleanup

`npm run cleanup:demo-financial-data -w @wilms/domain`

7. Cron and scheduler

| Job | Schedule | Endpoint |

|----|-----|-----|

| Notification dispatch | 06:00 UTC daily | `/api/cron/notifications` |

Configured in `vercel.json`. Verify cron execution in Vercel dashboard logs.

8. User management operations

Invite new user

Super Admin! Settings! Users! Invite. User receives email with invitation link. User clicks link and is INVITED to ACTIVE.

Force logout

Super Admin! User profile! Force logout. Invalidates all active sessions for user.

Permission overrides

Grant individual permissions beyond role defaults. All overrides audited.

Password reset

Users can self-service via forgot-password flow. Admin cannot view passwords.

9. Incident response

Severity levels

Level	Description	Response time
P1	Production down, data integrity risk	Immediate
P2	Major feature degraded	4 hours
P3	Minor issue, workaround available	24 hours
P4	Cosmetic, documentation	Next sprint

P1 procedure

1. Create ops incident in WILMS. 2. Check health endpoint and Vercel status. 3. Check Neon status page. 4. Rollback deployment if recent deploy caused issue. 5. Notify programme director. 6. Post-incident review within 48 hours.

10. Maintenance windows

Schedule via Ops module before:

- Database migrations with downtime risk
- Major version deployments
- Neon maintenance periods

Maintenance window displays banner to all users.

11. Security operations

Task	Procedure
Rotate session secret	Update WILMS_SESSION_SECRET in Vercel; all users re-login

| Rotate production users | npm run rotate-production-users -w @wilms/domain |

| Review failed login attempts | Audit log filter by auth.login_failed |

| Upload audit | Review upload logs for anomalous file types |

12. Performance monitoring

- Bundle budget: npm run bundle:budget-check
- Performance budget: npm run perf:budget-check
- API rate limit: 300 req/min global; monitor 429 responses

13. Documentation maintenance

Generate updated PDF/DOCX after documentation changes:

npm run docs:generate

Outputs in documentation/pdf/ and documentation/docx/.

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